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Modelling Economic Policy Issues

How financial development shapes globalization's impact on income inequality in Asia?

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ABSTRACT

Globalization in the form of increased trade, technological advancements, and capital flows has spurred economic growth and lifted millions out of poverty across Asia, but these advantages have not been equally distributed, leading to income inequality. While previous research has highlighted globalization's role in income inequality, little attention has been given to the moderating role of financial development. As Asian economies become more integrated into the global system, understanding how financial development mediates effects of globalization on income disparity has become crucial for policymakers, researchers, and practitioners. This study empirically investigates how financial development shapes the impact of globalization (trade, technological, and financial) on income inequality in Asian countries. Using a fixed effect panel data model with data for 22 Asian countries, we show that while all three modes of globalization—trade, technological and financial—aggravate the income gap, the influence of these types of globalization on income inequality is reduced when financial development takes place. In particular, the study finds that if a country is financially developed, the net effect of all types of globalization on income inequality is negative, meaning that in a financially developed country, globalization may actually help reduce income inequality. In response to these findings, we suggest policy recommendations, including increasing access to banking services and promoting financial literacy for financial inclusion, prioritizing sectors with high job creation potential, and providing reskilling support for globalization.

1. Introduction

Globalization can be interpreted as the connectivity and interdependence of people, economics, cultures, and nations worldwide (Brady et al., 2005). While the idea of interconnection is not new, the word 'globalization' has become more common in recent years due to trade liberalization, technology improvements, and communication innovations that have sped up the rate at which the world is coming together (Jaumotte et al., 2013; Bong and Premaratne, 2019). These developments have notable effects, including economic growth, changes in trade patterns, greater capital mobility, and technological progress (Berg et al., 2018).

Globalization is changing our connected world, bringing both opportunities and challenges. On one side, more trade, technological progress, and the flow of capital have helped economies grow, getting many people out of poverty (Cabral et al., 2016). But on the other side, not everyone has benefited the same from this global connection, which has led to an increase in income disparities (Firebaugh and Goesling, 2004; Autor et al., 2008; Mills, 2009). Income inequality is a term that refers to uneven distribution of

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income among people in a society, as a result of which, gaps between the rich and the poor becomes significant, resulting in socio-economic disparities (Asteriou et al., 2014).

This study identifies several ways in which income inequality is impacted by globalization. Trade globalization (Bensidoun et al., 2011; Rodríguez-Pose, 2012), for instance, entails the opening of markets and liberalization of trade and policies, which in turn empowers countries to specialize and engage in international trade. This has resulted in economic growth and job creation on one hand while wage polarization on the other. Export-oriented industries often pay higher wages to their highly skilled workers while low-skilled workers are likely to lose jobs or experience no wage growth in less competitive sectors, resulting in wage disparities (Zhu et al., 2005). The integration of financial markets around the world and the increase in cross-border flow of capital, known as financial globalization, have led to new economic opportunities and investment. Nonetheless, many people are not able to participate in financial globalization because financial markets and services are mainly accessible by wealthy people, hence leading to an increasing gap between rich and poor communities (Jaumotte et al., 2013; Bumann and Lensink, 2016). The last one, technological globalization, driven by many everyday developments, has transformed markets and economies in several countries and has improved productivity and output growth. However, such 'digitization' encompassing technological developments usually requires specialized skills (Giri et al., 2021) that increases the demand for skilled labor, whereas it reduces job opportunities for unskilled workers. This technological gap is considered another factor in increasing socioeconomic disparities.

Asia, for the last few decades, has been one of the greatest examples of globalization as well. Asia is a highly diverse region, encompassing economies at various stages of development, from advanced economies like Japan and South Korea to emerging markets such as China and India. During the second half of the 20th century and up until the present day, export-led strategies, combined with FDI, have underpinned growth and industrialisation across much of Asia (Munir and Bukhari, 2020). The region turned into a global player, pushing for integration by way of trade agreements and opening up of financial markets to the whole of the global system. Crucially, technological developments — especially in countries such as South Korea and Japan — have been very significant. Asia received huge foreign investments, which resulted in growth of manufacturing, technology, and services industries. However, while globalization has spurred economic growth and lifted millions out of poverty across Asia, but these advantages have not been equally distributed, leading to income inequality (Wong, 2016; Park, 2017).

Looking at Tables A1 and A2 in the appendix, we find that the trade globalization indicator (trade to GDP ratio) and the income inequality index (Gini coefficient) have followed similar trends in many Asian countries. Cyprus, for example, has observed a significant increase in trade globalization from 111.3 to 151.9 between 2007 and 2019. However, during the same period, it also experienced an increase in income inequality from 48.3 to 49.1. A similar trend is observed for Japan, where the trade globalization indicator increased from 32.8 to 35.2, while the income inequality index rose from 45.8 to 48.9 during the same period. Korea also shows a comparable pattern: trade globalization increased from 73.9 to 75.8, and income inequality rose from 34.3 to 37 between 2007 and 2019. Countries like Malasia and India, have both witnessed decline in their participation in trade globalization and income inequality levels during 2007–2019, suggesting that trade globalization did not contribute to reducing income inequality in these countries. Mongolia and Turkey, despite witnessing significant increases in trade globalization, did not observe any decline in their income inequality levels. On similar lines, Armenia and Vietnam, which also observed a significant increase in trade globalization, have achieved only a slight decline in their inequality index. Overall, the data suggests that while some nations have made progress in reducing income inequality, the majority of Asian countries have not experienced significant reductions in income inequality despite increased participation in the globalization process. This highlights the need to study the complex relationship between globalization and income inequality in the context of Asia.

In this globalization era, the question of how does financial development influence (Mishkin, 2006) the effect of globalization on income inequality has drawn vast attention among Asian countries, who are lately being emerged into world financial system. As part of our broader work on globalization and income inequality in Asia, we investigate the interconnected relationship between globalization and financial development. Financial development serves as the bedrock for the efficient allocation of resources, investment, and economic growth (Banerjee and Newman, 1991; Galor and Zeira, 1993; Aghion and Bolton, 1997). Still, its engagement with international trade, investment, and capital flows complicates the situation concerning the income distribution. This relationship dynamics is important to weigh in for policymakers, researchers, and practitioners who are looking for ways to solve the issue of economic inequality, particularly with the Asian context. In this study, we will focus on this complex relationship and seek to examine how the level of financial development within Asian economies shapes the effect of globalization on income distribution.

The growing importance of financial markets and financial development and their implications on income distribution among Asian countries provide motivation to conduct this research. Even though globalization promote economic growth, the effects of globalization on the income distribution are more complicated, and thus requires more detailed analyses (Mills, 2009). The function of financial development as a mediator in this relationship is under explored, and a thorough knowledge is required for informed policy decisions.

The following are the study's main literary contributions: First, this study examines the unrecognized role of financial development. While many studies have looked at trade, technological, and financial globalization separately, one important factor is often overlooked: financial development. It is noted that existing studies frequently miss how a country's level of financial development affects the globalization's impact on income inequality. The present study seeks to fill this gap by first classifying nations based on their financial development and then analysing how a nation's financial development influences the specific effects of globalization on income inequality. Second, while examining the wide range of globalization components, most studies have concentrated on trade and financial globalization while ignoring the transformative impact of technological globalization. There have been important technological advancements and implications since 2014. Some major breakthroughs include the development of smart gadgets, increased automation in many industries, advancements in healthcare technology, the expansion of digital platforms, and the continued evolution of communication technologies. The technological landscape is always changing (Gravina and Lanzafame, 2021), and new inventions continue to alter how we live, work, and interact. In today's environment, when technology is constantly advancing, this creates a substantial gap in our understanding of how it influences income distribution. This study addresses this gap by incorporating all three forms of globalization (trade, financial and technological) while looking at how a country's financial development shapes the impact of globalization on income inequality in Asian countries, hence ensuring a more comprehensive analysis of the subject.

Here is the remainder of the study: The literature review on trade, financial, technological globalization, and their connections to income disparity is presented in Section 2. In Section 3, the model, approach, and data are covered. Findings of empirical analysis is presented in Sections 4 and 5 discusses the conclusion and policy suggestions.

2. Literature review

Globalization, in the form of greater trade openness, technology/knowledge transfers, and increased foreign capital flows has significantly changed the economic landscape of Asia and the world. There has been a significant focus on the link between globalization and income inequality in recent research. Heimberger (2020), SenGupta (2021), and Gravina and Lanzafame (2021) noted that globalization augments income inequality. Similarly, Huh and Park (2021) highlighted that regional integration further amplifies this disparity. Atif (2012) also found that globalization intensifies income inequality, particularly in developed countries, while SenGupta (2021) observes a similar effect in developing nations. Although globalization is viewed as a potential source of economic growth and development, its effect on income inequality is still under debate with critics of globalization arguing that the economic benefits brought about by globalization is regressive rather than progressive so that they are not extended to all the population but more to a selective group of individuals or groups within one particular country. Globalization is often linked to inequality but with often divergent and contrary results (Mills, 2009).

2.1. *Impact of trade globalization*

Globalisation which is basically a liberalization of international movement of resources has elicited various discussions among various researchers as to its impacts on incomes. Regarding the discussion about trade globalization's impact on income disparities, there are two main positions among scholars (Munir and Bukhari, 2020). The first suggests that increased international trade reduces income inequality by promoting economic growth, fostering specialization, and enhancing efficiency, which raises living standards and narrows income gaps (Hui and Bhaumik, 2023; Rimidis and Butkus, 2023). In addition, according to this view, trade globalization, through comparative advantage, enhances productivity and wages, thus reducing income inequality (Giri et al., 2021; Chakrabarti, 2000). This view is also supported by studies such as Rudra (2004), Anderson (2005), Georgantopoulos and Tsamis (2011), and Jaumotte et al. (2013), suggesting that openness of trade can reduce inequality, especially when accompanied with education and social investments in underdeveloped countries.

Under the second view, however, it is suggested that globalization of trade contributes to greater inequality in income since its benefits are tilted towards capital- and skill-intensive industries and workers, while labour-intensive industries and low-skilled labors suffer a negative outcome (Lee, 2006; Sethi et al., 2021). Opponents of globalization point out that this benefits only the large corporations and rich people, increasing income inequality especially in developing countries (Silva and Leichenko, 2004). Other studies like Beckfield (2006) and Munir and Sultan (2017) also showed that globalization deepens income inequality. This is especially the case in regions characterized by vast differences in resource endowments, population size and trade costs. Similarly, Spilimbergo et al. (1999) and Milanovic (2005) suggested that so-called trade globalization mainly benefits the already rich, increasing the income inequality.

2.2. *Impact of technological globalization*

This section of the literature reviews the impacts of technological globalization on income inequality. In terms of modern global economy, technological globalization is defined as the spread of technology and information across borders. Some scholars argue that income inequality in 21st century has been a result of globalization, particularly technological globalization which has heightened the value of human capital. This underscores the importance of both education and skills in reducing this gap between the rich and poor. Autor et al. (2008) supported this argument by examining how the integration of technology into American labor markets led to the emergence of wage differentials and thus concluded that income inequality worsened due to technological globalisation. Jaumotte et al. (2013) found similar results in their study, showing that technological globalization has more severe effects on income inequality than globalization alone. It is suggested that surging technological globalization has increased the demand for skills and education, often leading to unequal income distribution that favors those with greater level of skills and education. This has sparked significant attention from researchers, resulting in mixed results across different country groups. Giri et al. (2021) while studying the impact of technological progress on income inequality in India, suggested that technological progress are inherently skill-based, favors skilled labor and capital-intensive production, thereby widens the income gap among low and high skilled labor.

A number of authors, however, suggest that the current process of globalization promotes technological expansion with considerable scope for innovation, productivity growth and economic progress, and hence can results in reduced income inequality. According to them, technological advancements increase productivity, creates far more jobs than it destroys, improves access to education, ultimately foster social mobility and reduce economic disparities. For example, Munir and Bukhari (2020) argued that technological globalization plays a major role in decreasing income inequality. The findings by Acemoglu and Autor (2011) claimed that while there was a positive relationship between technological innovation and economic growth, its impact on income inequality depended on skill biased technical change, education policies and labor market institutions. Another study by Bessen (2019) which looked at how technological progress affects income inequality, found that even though the technological progress causes a temporary upsetting in employment and income distribution, it has also increased productivity and living standards for different strata of society over time.

2.3. Impact of financial globalization

There has been a recognition of the fact that over the past few decades, financial globalization characterized by integration of national financial markets and institutions with their global counterparts is an essential feature in the current world economy. Therefore, this section on literature review concerns about examining how income inequality is affected by financial globalization or financial interconnectedness among nations. For instance, Jaumotte et al. (2013) assessed whether or not income inequality has been influenced by financial globalization in 51 countries from 1981 to 2003, where they noted that most countries experienced increased income inequality with the increase in financial globalization. Munir and Bukhari (2020) also observed that Asian countries' rise in income inequalities are linked with their financial integration implying unevenly distributed benefits of financial globalization. Similarly, Giri et al. (2021) discovered that financial globalization, along with trade and technological advancements, exacerbated income imbalances in India from 1982 to 2018.

LEE (2014) while analysing the impact of financial integration, on income distribution and poverty and concluded that financial globalization exacerbates income disparity and poverty. Similarly, Khan et al. (2021) found that greater openness to international capital flows accelerated the rise of income inequality globally. Contrarily, some studies present opposing views. Milanovic (2005) studied the relationship between openness and wealth distribution in 95 low, middle, and high-income countries, finding that while trade worsened income disparity, while financial stability and integration reduced it. Delis et al. (2014) found that financial globalization reduces income inequality, while Agnello et al. (2012) deemed its impact insignificant. Bumann and Lensink (2016) also argued for an income inequality-reducing effect of financial globalization.

2.4. Role of financial development

In conclusion, a variety of viewpoints on the connection between globalization and income inequality are presented in the literature, with a focus on the complex roles played by technology and financial factors. While some studies highlight exacerbating effects, others underscore the potential for globalization, particularly in trade, to alleviate income inequality. There is a notable gap in the empirical literature regarding how a country's financial development (such as improving banking, credit access, and stronger financial markets) influences the impact of globalization on income inequality. Existing studies (Mishkin, 2006; Bittencourt et al., 2019; Sethi et al., 2021) suggests that financial development by improving allocation of resources, reducing credit constraint, access to financial services will help the poor and reduce income inequality. Therefore, globalization in higher financially developed countries can possibly improve the outcomes of globalization. The existing research landscape has not sufficiently delved into the intricate dynamics between financial development and the consequences of globalization for income distribution. This void underscore the need for comprehensive empirical investigations to unveil the nuanced interactions and potential mediating role of financial systems in shaping the effect of globalization on income inequality.

Given the growing importance of financial development, this research aims to explore how financial development shapes globalization's impact on income inequality in Asian countries? As observed in the literature, while globalisation of various types may have mixed effects on the income inequality depending on the circumstances, we expect that the financial development of a country can help improve the outcome associated with globalisation's impact on income inequality. This study, therefore, aims to explore following hypothesis.

H1: Financial development helps a country achieve a better outcome with respect to the impact of trade globalization on income inequality
H2: Financial development helps a country achieve a better outcome with respect to the impact of technological globalization on income inequality
H3: Financial development helps a country achieve a better outcome with respect to the impact of financial globalization on income inequality

2.5. Control variables

This analysis incorporates urbanization, GDP per capita, R&D expenditure as control variables to consider their effects on income disparity (Munir and Bukhari, 2020; SenGupta, 2021; Tabash et al., 2024) along with different forms of globalization. A general conventional assumption is that urbanization fuels income disparities because of the prevailing economic pattern whereby cities create huge gaps in income distribution across classes. Typically, during city growth phases, the centers of economic activities are established where few individuals benefit from industries associated with urbanization such like technology or services that attract high wages but do not employ many low-skilled workers. In addition, this may increase income inequality between rural regions and those residing within cities resulting into a dualistic society. Conversely one expects that increases in GDP per capita will reduce inequality between groups. A higher level of GDP per capita implies significant development potentials as well as improved living conditions, translating into more egalitarian distribution of earnings throughout a population. Similarly, it is generally assumed that higher level of R&D spending will reduce income inequality. A higher level of investment in R&D promotes innovation, which can boost productivity and create better-paying jobs. Thus, while urbanization may lead to greater proportions of the population with small earnings, higher gross domestic product per capita and greater investment in R&D implies a fairer income allocation (Bergh et al., 2008; Villanthenkodath et al., 2024; Tabash et al., 2024).

3. Data, methodology and empirical model

The current section describes the data source and definition, discusses the methodology applied for estimating the empirical model.

3.1. Data: source and definition

This study employs yearly data from 2007 to 2019 for 22 Asian countries[1,2] to test our hypotheses regarding how financial development mediates effects of different forms of globalization on income disparity. The selection of these countries was driven by the availability of reliable, consistent, and comprehensive data on key variables, including income inequality, financial development, globalization indicators and control variables, over the study period. Data are primarily sourced from the World Development Indicators (WDI), World Bank and the IMF database. The World Bank database provides detailed information on individual countries sourced from officially recognized international sources. It includes data on country's trade, capital inflows and technology, GDP per capita, R&D expenditure, and urban population. Our sample comprises countries from the Asia. Income inequality (INQ) is quantified using the Gini coefficient, the most widely utilized index for measuring income inequality (Foster et al., 2013, p. 13). We obtain the income inequality measures from the Standardized World Income Inequality Database (SWIID) by Solt (2020) (Sethi et al., 2021). Trade globalization is measured by trade to GDP ratio i.e. as the sum of exports and imports divided by GDP, while financial globalization is measured as net private capital flows divided by GDP. Technological globalization is calculated as high-tech export divided by GDP. The analysis incorporates additional control variables in accordance with Munir and Bukhari (2020), SenGupta (2021) and Tabash et al. (2024) i.e. GDP per capita, urban population and research and development (R&D) expenditure. In this study, we introduce a dummy variable, 'DUMMY_FinDevt,' which takes a value of 1 if a country' financial development index is greater than 0.7, defining that country as financially developed. Conversely, if a country's financial development index is less than 0.7, the dummy variable takes a value of 0, indicating that the country is financially underdeveloped. The data on financial development index[3] is obtained from IMF Database. Table 1 summarizes the dependent, independent and control variables used in the study.

3.2. Empirical model

Our empirical models estimate the effects of three distinct forms of globalization—trade, technological, and financial—on income inequality separately. Additionally, they seek to understand how these forms of globalization interact with financial development to influence income inequality in Asian countries. As discussed earlier, we construct a dummy variable, 'DUMMY_FinDevt,' which is assigned a value of 1 for financially developed countries, and 0 otherwise. To capture the combined impact of financial development and globalization, we introduce an interaction term. To assess the effects of different dimensions of globalization [i.e. technological globalization, trade globalization, and financial globalization] and their interaction with financial development on income inequality, we estimate the following three models:

Model 1: $LINQ_{it} = \beta_0 + \beta_1\, LTradeG_{it} + \beta_2\, (LTradeG_{it}{*}DUMMY\_FinDevt) + \beta_3\, LGDPPC_{it} + \beta_4\, LURBN_{it} + \beta_5\, LRDEX_{it} + \varepsilon_{it}$
Model 2: $LINQ_{it} = \alpha_0 + \alpha_1\, LTechG_{it} + \alpha_2\, (LTechG_{it}{*}DUMMY\_FinDevt) + \alpha_3\, LGDPPC_{it} + \alpha_4\, LURBN_{it} + \alpha_5\, LRDEX_{it} + \varepsilon_{it}$
Model 3: $LINQ_{it} = \gamma_0 + \gamma_1\, FinG_{it} + \gamma_2\, (FinG_{it}{*}DUMMY\_FinDevt) + \gamma_3\, LGDPPC_{it} + \gamma_4\, LURBN_{it} + \gamma_5 LRDEX_{it} + \varepsilon_{it}$

Where LINQ denotes income inequality, while LTradeG represents trade openness and is used to measure trade globalization. DUMMY_FinDevt is a dummy variable which is assigned a value of 1 for financially developed countries and 0 otherwise. LTechG stands for technological globalization, and FinG indicates financial globalization. Additionally, three control variables are included:

[1] The study focuses on Asia for several key reasons. First, the region's diversity-ranging from developed economies like Japan and South Korea to emerging economies such as China and India-provides a robust basis for understanding how interaction of globalization with financial development influence income inequality at different economic levels. Second, Asia, for the last few decades, has been one the greatest examples of globalization, with export-led strategies, trade agreements, FDI, and technological advancements driving industrialization and economic growth (Munir and Bukhari, 2020), making it an important region to study the impact of globalization on income inequality.

[2] List of 22 Asian countries: Armenia, China, Cyprus, Georgia, India, Indonesia, Israel, Japan, Jordan, Kazakhstan, Korea, Kyrgyz Republic, Lao PDR, Malaysia, Mongolia, Pakistan, Saudi Arabia, Singapore, Sri Lanka, Thailand, Turkiey, and Vietnam.

[3] The Financial Development Index captures multidimensional aspects of financial development by measuring how developed financial institutions and markets are in terms of depth (size and liquidity), access (the ability of individuals and companies to access financial services), and efficiency (the capacity to provide services at low cost with sustainable revenues and the level of activity of capital markets). The index is constructed by aggregating six lower-level sub-indices, which measure the depth, access, and efficiency of financial institutions and markets, into two higher-level sub-indices, Financial Institutions and Financial Markets, and ultimately aggregating them to form the overall Financial Development Index (Sahay et al., 2015).

Table 1
Definition and source of variables.

Variable Name	Measurement	Data Source
Income Inequality (INQ)	Gini Coefficient	Standardized World Income Inequality Database (SWIID)
Financial Development Index (FinDIndex)	Financial Development Index	IMF Database
Dummy for Financial Development (`DUMMY_FinDevt)	Takes a value of 1, if FinDIndex is greater than 0.7, and 0 otherwise	Constructed
Trade Globalization (TradeG)	Sum of exports and imports as a share of GDP	WDI, World Bank
Technological Globalization (TechG)	High Tech Export as a share of GDP	WDI, World Bank
Financial Globalization (FinG)	Net Private Capital Flows as a share of GDP	WDI, World Bank
GDP Per Capita (GDPPC)	Total GDP divided by total population	WDI, World Bank
Urbanization (URBN)	Urban population as a share of total Population	WDI, World Bank
R&D Expenditure (RDEX)	R&D Expenditure as a share of GDP	WDI, World Bank

LGDPPC, which stands for GDP per capita, LRDEX, which represents R&D expenditure as a share of GDP, and LURBN, which represents urbanization. All variables, except the dummy and financial globalization, are taken in natural logarithm, as indicated by the prefix L.

3.3. Methodology

The study makes use of a panel data methodology to examine how globalization affects income distribution. Panel data offers several edges over time-series and cross-sectional data, including control for individual heterogeneity, a larger number of data points, greater degrees of freedom, and a reduction in collinearity between independent variables, thus enhancing the accuracy of econometric estimates. Panel data can be used in a number of ways to lessen the severity of significant econometric issues that frequently occur in empirical research (Hsiao, 2022). When studying more complex behavioral models, panel data can more accurately identify and quantify effects that are difficult to discover in pure cross-section and time-series data (Baltagi, 2008; Wooldridge, 2010).

In this situation, straightforward estimate methods such as pooled OLS are probably going to be ineffective because they do not account for and compensate for the unobservable individual country effects. Given the panel dataset structure, which encompasses observations across multiple countries over time, employing a Fixed Effects (FE) model holds notable advantages for analyzing the relationship between globalization and income inequality (Plümper and Troeger, 2007; SenGupta, 2021). FE models are particularly well-suited for addressing the nuances inherent in panel data analysis, especially when dealing with unobserved time-invariant country-specific factors (Mundlak, 1978; Wooldridge, 2010). This is to say that FE models use country-specific intercepts so as to control for possible biases that arise from unobserved variables which have not changed within each country over time (Baltagi, 2008). This feature is crucial for capturing the heterogeneity across countries and preserving temporal variation within each country over the study period. Additionally, FE models allow for the identification of country-specific trends or shocks, thus enabling a nuanced understanding of the factors driving income inequality dynamics within individual countries. Overall, given these strengths, employing a Fixed Effects model emerges as a suitable and robust approach for investigating the complex relation between globalization and income disparity within a panel dataset framework.

4. Results and discussion

Table 2 indicates the dataset comprises 286 observations detailing various economic and social indicators. Among these, the Gini coefficient stands out as a measure of income inequality among the population, averaging approximately 43.18 with values ranging from 34.2 to 52.3. The Financial Development Index, reflecting the maturity of financial systems, has a mean value of around 0.446, spanning from 0.082 to 0.908. Technological Globalization, capturing the diffusion of technology across countries, averages at about 0.056, ranging from 0.0003 to 0.6386. Trade Globalization, quantifying the degree of global trade integration, shows a mean value of approximately 0.954, with a wide range from 0.244 to 4.373. Financial Globalization, indicating the global integration of financial markets, has an average value of around 0.103, fluctuating from −0.372 to 2.801. GDP Per Capita, a key metric of economic output per person, stands at a mean of $20,714.64, varying from $2769.43 to $102,630.90. The proportion of Urban Population, representing the share of people living in urban areas, averages around 0.604, ranging from 0.182 to 1. Lastly, R&D expenditure as a percentage of GDP

Table 2
Descriptive statistics of variables.

Variables	Obs.	Mean	Std. Dev.	Minimum	Maximum
Income Inequality (INQ)	286	43.18	5.15	34.20	52.30
Financial Development Index (FinDIndex)	286	0.4459	0.2146	0.0821	0.9084
Technological Globalization (TechG)	286	0.0563	0.1121	0.0003	0.6385
Trade Globalization (TradeG)	286	0.9542	0.6824	0.2439	4.37
Financial Globalization (FinG)	286	0.1027	0.3227	−0.3717	2.80
GDP Per Capita (GDPPC)	286	20,714.64	19,089.81	2769.43	102,630.90
Urbanization (URBN)	286	0.6042	0.2285	0.1819	1
R&D Expenditure (RDEX)	286	0.01028	0.01258	0.00038	0.05215

has a mean of 0.01 and ranges from 0.00 to 0.05. These statistics paint a comprehensive picture of the economic and social landscape across the observations, offering insights into income distribution, financial development, globalization dynamics, and urbanization trends.

4.1. Panel results of Model 1, Model 2 and Model 3

Table 3 shows the results of Model 1, examining the impact of trade globalization on income inequality in Asian countries. We note that trade globalization has a positive coefficient, suggesting that rising levels of trade globalization cause rising levels of income disparity. This outcome is consistent with previous research by Lee (2006), Ali and Isse (2007), Hepenstrick and Tarasov (2015), Munir and Sultan (2017) and Sethi et al. (2021). A possible interpretation of why trade liberalization has a harmful effect on income distribution is the fact that trade globalization has driven considerable skill-biased entity-induced technical change in industries most significantly affected by international competition and this is the place where much of the increase in relative demand for skilled labour seems to originate. In these sectors, high-skilled workers tend to earn higher wages due to their specialized skills and ability to adapt to new technologies and market demands (Zhu et al., 2005). By contrast workers with lower skills in relatively less-competitive sectors will end up in jobs where their wages are pushed or even join the dole queue as they simply cannot compete with lower-cost imports.

Interaction term of trade globalization with financial development has a negative coefficient, suggesting that financial development mitigates the impact of trade globalization on income inequality, which verifies the first hypothesis (H1). We also note that this negative coefficient, in absolute term, is greater than the positive coefficient associated with trade globalization, implying that although trade globalization has a positive influence on income inequality but if the country is financially developed, the net effect of trade globalization on income inequality is negative, meaning in a financially developed country, trade globalization can reduce income inequality. This indicates that enhanced financial development can address income inequality by improving access to financial services (Sethi et al., 2021) for individuals and businesses across income groups. This empowerment enables marginalized populations to invest in education and assets, helping them reaping the benefits of trade globalization, thereby reducing income disparities. Financial development also fosters investment in human capital, enhancing labor productivity and employability, while supporting SMEs stimulate entrepreneurship, job creation and participate in trade activities, contributing to inclusive growth and narrowing income differentials.

Among controls, GDP per capita negatively and significantly impacts income inequality, while urbanization positively impacts the income inequality. The findings align with the expected outcomes: urbanization has been associated with an increase in income inequality, likely due to the concentration of wealth and opportunities in urban areas, which often exacerbates wage gaps between skilled and unskilled workers (SenGupta, 2021). Conversely, the rise in GDP per capita appears to mitigate this effect, contributing to a reduction in income inequality by promoting broader economic growth and fostering a more equitable distribution of income across different segments of the population. A higher R&D expenditure also leads to fall in income inequality as higher level of investment in R&D promotes innovation, which can boost productivity and create better-paying jobs.

Table 4, which represents the results of Model 2, shows a positive coefficient for technological globalization, suggesting that rising levels of technological globalization cause rising levels of income disparity. This outcome is consistent with previous research by Jaumotte et al. (2013), Gravina and Lanzafame (2021), and Giri et al. (2021). One explanation for this result can be that, technological globalization, propelled by rapid advancements in technology, has led to significant transformations across industries enhancing productivity and efficiency. However, the adoption of new technologies often necessitates specialized skills (Autor et al., 2008; Giri et al., 2021) creating a disparity in the labor market. Skilled workers, equipped to navigate and leverage these technologies, find themselves in high demand, commanding higher wages and enjoying greater job security. Meanwhile, those lacking these specialized skills, typically low-skilled or unskilled workers, face diminished opportunities and lower wages as their roles become increasingly automated or outsourced. This phenomenon exacerbates income inequality, widening the gap between those who can adapt to technological changes and those left behind.

Interaction term of technological globalization with financial development has a negative coefficient, suggesting that financial development mitigates the impact of technological globalization on income inequality, which verifies the second hypothesis (H2). We

Table 3
Results of Model 1.

Dependent Variable: Log of Income Inequality (LINQ)			
Variable	**Coefficient**	**SE**	**p-value**
LTradeG	0.0263***	0.0093	0.005
(LTradeG)*(DUMMY_FinDevt)	−0.0591**	0.0276	0.034
LURBN	0.0774*	0.0460	0.094
LGDPPC	−0.0382***	0.0084	0.000
LRDEX	−0.0036	0.0026	0.161
_cons	4.1560***	0.1016	0.000
N	286		
R-sq: within = 0.1413, between = 0.0147, overall = 0.0109			
F(5,259) = 8.52 (p = 0.000)			

***, ** and * depict significance at 1 %, 5 % and 10 %.

Table 4
Results of Model 2.

Dependent Variable: Log of Income Inequality (LINQ)			
Variable	**Coefficient**	**SE**	**p-value**
LTechG	0.0125*	0.0068	0.066
(LTechG)*(DUMMY_FinDevt)	−0.0547*	0.0281	0.053
LURBN	0.0367	0.0439	0.405
LGDPPC	−0.0376***	0.0086	0.000
LRDEX	−0.0032	0.0026	0.228
_cons	4.1180***	0.1020	0.000
N	286		
R-sq: within = 0.1250, between = 0.0296, overall = 0.0259 F(5, 259) = 7.40 (p = 0.000)			

***, ** and * depict significance at 1 %, 5 % and 10 %.

also note that this negative coefficient, in absolute term, is greater than the positive coefficient associated with technological globalization, implying that although technological globalization has a positive influence on income inequality but if the country is financially developed, the net effect of technological globalization on income inequality is negative, meaning in a financially developed country, technological globalization can reduce income inequality. This also indicates that financially developed countries lower the effect of technological globalization on income inequality. A strong financial system benefits a nation in many ways. Primarily, it facilitates simpler loan acquisition, particularly for startups and small enterprises. This encourages innovation and job growth even in industries where technology is causing disruptions. Second, well-established financial markets make it easier to fund training and educational initiatives (Mishkin, 2006). This assists workers in modifying their skill sets to meet the demands of an evolving labor market due to technology improvements. Finally, by ensuring a greater proportion of the population takes part in the growth brought about by new technology, financial inclusion helps close the wealth gap. All things considered, a financially developed nation is better positioned to spread the advantages of technological globalization more broadly, reducing the potential for increased income gap. Among controls, similar to the previous model, GDP per capita and R&D expenditure negatively impacts income inequality, while the urban population positively impacts income inequality.

We obtain the similar results in Table 5, which shows the results of Model 3, examining the impact of financial globalization on income inequality in Asian countries. The coefficient for financial globalization is although insignificant but it is positive, suggesting that rising levels of financial globalization can lead to higher income disparity. This outcome is consistent with previous research by Figini and Gö rg, (2011), Jaumotte et al. (2013), Furceri and Loungani (2018), Khan et al. (2021), and Hui and Bhaumik (2023). Financial globalization, while often seen as a driver of economic growth, can contribute to rising income inequality through several key mechanisms. Companies can exploit financial globalization to offshore operations and profits to countries with lower tax rates, reducing tax revenue that could be used for social programs and infrastructure, thereby disproportionately benefiting wealthy corporations and shareholders. This leads to calls for quick profits, which translate into outsourcing of jobs by companies in the developed countries having a new supply of cheap capital through the individuals in the developing world and also through automation of jobs that were previously done by the lower skilled workers in the developed world (Khan et al., 2021). Some outcomes include that new jobs can also be generated; however, these jobs demand more sophisticated skills, thus disemploying some workers. Furthermore, financial globalization helps the rich to gain access to international assets, thereby deepening inequality through promoting access to financial markets by the rich (Jaumotte et al., 2013; Bumann and Lensink, 2016).

In addition, similar to the case of trade and technological globalization, the interaction term of financial globalization with financial development has a negative coefficient, suggesting that financial development mitigates the impact of financial globalization on income inequality, which verifies the third hypothesis (H3). This indicates that a country's financial development can help mitigate the negative impact of financial globalization on income inequality by better allocating resources (Mishkin, 2006; Sethi et al., 2021), and by enhancing access to financial services for a broader segment of the population. Financial development also allows smaller domestic enterprises and small investors to take the advantage of financial globalization by accessing international financial markets,

Table 5
Results of Model 3.

Dependent Variable: Log of Income Inequality (LINQ)			
Variable	**Coefficient**	**SE**	**p-value**
FinG	0.0055*	0.0032	0.086
(FinG)*(DUMMY_FinDevt)	−0.0077	0.0877	0.930
LURBN	0.0295	0.0443	0.506
LGDPPC	−0.0334***	0.0085	0.000
LRDEX	−0.0034	0.0026	0.205
_cons	4.0770***	0.1013	0.000
N	286		
R-sq: within = 0.1103, between = 0.1018, overall = 0.0845 F(5,259) = 6.42 (p = 0.000)			

***, ** and * depict significance at 1 %, 5 % and 10 %.

providing them with the opportunity of diversification, risk management, access to cheaper funds, and the potential to earn greater income, thereby helping to reduce disparities among people. Among the control variables, just as in the case of Model 1 and Model 2, we find that GDP per capita and R&D expenditure negatively impacts income inequality, while the urban population positively impacts income inequality.

We further disaggregated the financial globalization (FinG) variable into net portfolio flows and net banking flows to examine how these different types of financial flows impact income inequality and the role financial development plays in that relationship. The results are presented in Tables A3 and A4 in the Appendix. We note that, just like the results in the original Model 3 (Table 5), rising levels of financial globalization—defined both in terms of an increase in net portfolio flows and also in terms of an increase in net banking flows—tend to increase income inequality. However, in financially developed countries, net banking flows tend to exacerbate the impact of financial globalization on income inequality.

4.2. Panel results using a composite index of globalization

The study also constructed a composite index of globalization comprising of three competing globalization variables (trade, technological, and financial globalization), using the Department of Commerce method. In this method, the three globalization variables were assigned a weight equivalent to their share in the country's GDP and summed. The resulting value was then normalized by the country's total GDP to construct the index. Using this composite index of globalization, the panel results on how globalization impacts the income inequality in Asian countries are reported in Table 6. We note that the coefficient of the globalization index is positive and significant, suggesting that rising levels of globalization lead to increased income inequality. We also note that the interaction term of globalization index with financial development has a negative coefficient, suggesting that financial development mitigates the impact of globalization on income inequality. Furthermore, the absolute value of the negative interaction coefficient is greater than the positive coefficient of the globalization index, implying that although globalization has a positive influence on income inequality but if the country is financially developed, the net effect of globalization on income inequality is negative —meaning that in a financially developed country, globalization may actually help reduce income inequality. These findings are consistent with the results of the three models presented earlier.

4.3. Additional estimations of the three models

We also re-estimated the three models many times; first by considering an alternative methodology to address any potential endogeneity; second by considering an alternative measure of income inequality; and third, by diving our sample countries into two sub-groups: (a) low- and middle-income countries, and (b) high income countries. These re-estimations helped check the robustness of the results.

4.3.1. Results based on GMM method

To address any potential endogeneity arising from omitted variable bias, we also re-estimated the three models using the GMM method. The GMM results for the three models are presented in Tables A5–A7 in the appendix. We find that these results are similar to our original findings in Tables 3–5, showing that all three modes of globalization—trade, technological, and financial—increase income inequality. In addition, consistent with our original findings, we find that the impact of globalization when interacted with financial development is negative and is more than the positive impact of globalization on income inequality. This implies that if a country is financially developed, the net effect of all types of globalization on income inequality is negative, meaning that in a financially developed country, globalization can reduce income inequality.

4.3.2. Results based on an alternative measure of income inequality

We also re-estimated the three models using an alternative measure of income inequality, top to bottom income quantile ratio, to check for the robustness. The results of the three models with this measure are presented in Tables A8–A10 in the appendix. We again note that the results are similar to those in Tables 3–5, that is, all three types of globalization lead to an increase in income inequality;

Table 6
Panel results with a composite index of globalization.

Dependent Variable: Log of Income Inequality (LINQ)			
Variable	**Coefficient**	**SE**	**p-value**
LGIndex	0.0102***	0.0038	0.009
LGIndex*(DUMMY_FinDevt)	−0.0267*	0.0137	0.053
LURBN	0.0724	0.0458	0.116
LGDPPC	−0.0384***	0.0084	0.000
LRDEX	−0.00378	0.0026	0.151
_cons	4.153***	0.1019	0.000
N	286		
R-sq: within = 0.1380, between = 0.0159, overall = 0.0117 F(5,259) = 8.29 (p = 0.000)			

LGIndex: Log of Globalization Index; ***, ** and * depict significance at 1 %, 5 % and 10 %.

however, when a country is financially developed, the net effect of globalization on inequality becomes negative, thus providing further support to our original findings.

4.3.3. *Low- and middle-income countries vs high income countries*

To perform an additional sensitivity check, we also divided our sample countries into two sub-groups: (a) low- and middle-income countries, and (b) high income countries, and then re-estimated the three models for these sub-groups. Tables A11–A13 in appendix present the results for low- and middle-income countries, while the results for high income countries are presented in Tables A14–A16. We find that, except for the model with financial globalization for low- and middle-income countries, the results regarding the impact of different types of globalization on income inequality and its interaction with financial development are similar our original findings for both sets of countries, thus confirming the robustness to our original findings.

5. Conclusion and policy suggestions

This study investigates how different forms of globalization—trade, technological, and financial—impact income inequality in Asian countries, how those impacts are influenced when these forms of globalization interact with a country's financial development. In other words, the study seeks to know how financial development shapes globalization's impact on income inequality in Asian countries. This study employs fixed effect panel data methodology which accounts for unobservable country effects, to test hypotheses and derive empirical results.

The findings show that all three modes of globalization—trade globalization, technological globalization, and financial globalization—aggravate the income gap. This imply that trade globalization can lead to job displacement in certain industries and unequal wage gains favoring higher-skilled workers. Similarly, technological globalization often results in skill-biased technological changes, widening the income gap between those with advanced skills and those without. Financial globalization may exacerbate income inequality by providing unequal access to financial markets, benefiting capital owners more than labor.

This study has also provided empirical confirmation that the financial development of a country significantly reduces the impact of all types of globalization on income inequality. In particular, the study finds that if a country is financially developed, the net effect of all types of globalization on income inequality is negative, meaning that in a financially developed country, globalization may actually help reduce income inequality. By efficiently allocating resources, reducing credit constraints, and providing funds for investment in human capital, financial development plays a crucial role in mitigating the effects of globalization on income inequality. Among controls, the effect of GDP per capita on income inequality is found to be negative, implying that an increase in GDP per capita tends to reduce income inequality. The R&D expenditure is also found to have negative influence but the impact is insignificant. In contrast, urbanization has a positive coefficient, which reflects that income inequality tends to worsen with urbanization.

The study highlights the need for effective financial development of a nation to reduce the negative effects of globalization on income inequality and suggests actionable strategies to achieve this. First, strengthen financial inclusion by increasing the presence of banking channels in underserved areas, particularly rural areas, can lead to a greater degree of financial inclusion and help marginalized citizens engage more fully in economic activities. This includes setting specific targets, such as increasing the percentage of rural residents with access to formal banking services and promoting mobile banking through partnerships with telecom providers to reduce transaction costs. Financial literacy programs should also be introduced to equip individuals with the knowledge and tools needed to manage their finances effectively.

Second, to address workforce challenges posed by globalization, education and skill development initiatives should focus on reskilling workers from industries adversely affected by global trade dynamics. This can include subsidized vocational training programs in high-demand sectors such as renewable energy, technology services, and advanced manufacturing. Governments can partner with industries to incentivize retraining and offer financial support for displaced workers during their transition. Additionally, the overall globalization strategy should target sectors where jobs can be created and skills can be developed.

Third, globalization often benefits large corporations unequally. Providing financial and technical support to SMEs, such as low-interest loans or grants for adopting new technologies, can help these businesses compete globally by producing higher-quality products and create more local jobs. Finally, the implementation of a robust monitoring and evaluation framework is important to analyze the success of these initiatives. Regular reporting and transparent assessments can ensure accountability and enable policymakers to refine strategies based on measurable outcomes. The goal of these strategies is to create an economic environment that is fairer and benefits all members of the society. With this, globalization can lead to more inclusive and equitable economic growth, benefiting both individuals and society as a whole.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work, the authors used Grammarly & Chatgpt in order to check for grammar, spelling, improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Virender Kumar: Writing – review & editing, Validation, Supervision, Project administration, Methodology, Conceptualization. **Simran:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Table A1
Trade globalization (Trade Volume as a percentage of GDP).

	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Armenia	57.0	54.5	57.3	64.6	69.4	76.0	77.6	75.8	71.7	76.1	87.2	92.5	96.1
China	62.2	57.6	45.2	50.7	50.7	48.3	46.7	44.9	39.5	36.9	37.6	37.6	35.9
Cyprus	111.3	112.9	102.8	109.2	110.9	112.7	121.2	131.2	137.7	139.6	148.5	149.0	151.9
Georgia	88.4	86.3	78.2	82.9	87.5	91.9	95.6	96.8	98.8	96.8	104.0	111.8	118.6
India	45.7	53.4	46.3	49.3	55.6	55.8	53.8	48.9	41.9	40.1	40.7	43.6	39.9
Indonesia	54.8	58.6	45.5	46.7	50.2	49.6	48.6	48.1	41.9	37.4	39.4	43.1	37.6
Israel	32.8	34.1	62.6	28.5	30.2	70.7	34.0	37.4	59.2	31.3	34.4	36.6	35.2
Japan	32.8	34.1	24.4	28.5	30.2	30.5	34.0	37.4	35.4	31.3	34.4	36.6	35.2
Jordan	146.0	140.9	112.2	114.2	118.7	117.9	111.5	109.9	95.4	88.7	90.1	88.0	85.8
Kazakhstan	92.2	94.3	75.8	74.1	73.1	73.7	65.4	65.0	53.0	60.3	56.8	63.5	64.9
Korea	73.9	95.5	86.1	91.4	105.6	105.5	98.0	90.6	79.1	73.6	77.1	79.0	75.8
K.Republic	137.1	146.1	133.4	133.2	136.2	139.7	134.0	125.1	111.0	105.8	100.6	98.9	79.2
Lao PDR	79.2	81.8	76.9	84.7	91.7	98.2	98.2	99.1	85.8	75.1	75.1	75.1	75.1
Malaysia	192.5	176.7	162.6	157.9	154.9	147.8	142.7	138.3	131.4	126.9	133.2	130.4	123.0
Mongolia	117.9	121.2	107.8	103.4	127.0	109.6	100.3	109.3	89.7	101.1	115.9	126.4	124.4
Pakistan	30.8	34.3	33.3	32.0	32.4	31.3	30.9	29.5	26.7	24.7	25.5	27.6	28.9
S.Arabia	94.9	96.1	84.9	82.5	84.9	82.9	81.9	79.6	69.5	59.9	61.8	62.0	60.2
Singapore	394.3	437.3	358.2	369.7	379.1	369.2	367.0	360.5	329.5	303.1	316.5	325.2	321.7
Sri Lanka	68.6	63.4	358.2	49.1	48.1	369.2	47.1	46.1	329.5	46.5	47.1	49.8	49.4
Thailand	129.9	140.4	119.3	127.3	139.7	137.7	132.5	130.9	124.8	120.6	120.9	120.8	109.7
Turkiey	47.9	50.5	46.8	46.7	53.3	52.8	52.5	53.8	51.1	48.3	55.8	62.6	63.2
Vietnam	154.6	154.3	134.7	114.0	125.3	123.2	130.8	135.4	144.9	145.4	161.0	164.7	164.7

Source: World Bank.

Table A2
Income inequality (Gini coefficient).

	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Armenia	48.4	48	47.6	47.5	47.3	47.5	47.7	47.9	48	48.1	48	47.7	47.3
China	47.3	47.9	48.2	48.6	49.1	49.1	49.2	49.1	48.9	48.9	49.1	49.4	49
Cyprus	48.3	48.5	48.6	48.7	48.9	49	49.1	49.2	49.2	49.2	49.2	49.1	49.1
Georgia	51.4	51.8	52.3	52.1	51.5	50.6	49.9	49.4	49	48.6	48.4	47.8	47.4
India	48.2	48.4	48.8	49.4	50.1	48.8	47.7	46.9	46.4	45.8	45.5	45.2	45.1
Indonesia	40.4	40.7	41.1	41.5	41.9	42.2	42.4	42.5	42.5	42.5	42.4	42.3	42.3
Israel	51.9	51.7	51.4	51	50.3	49.6	48.7	48.1	47.6	47.1	46.9	47	47.4
Japan	45.8	46.8	47.2	47.8	48	48.2	48.3	48.2	48.5	48.6	48.8	48.9	48.9
Jordan	39.3	39.1	39.1	39	39	39	39	39	39.1	39.2	39.2	39.2	39.2
Kazakhstan	37.7	37.2	36.8	36.5	36.2	36	35.8	35.6	35.6	35.5	35.6	35.6	35.7
Korea	34.3	34.4	34.6	34.5	34.8	34.5	34.8	34.6	35.1	35.7	36.5	36.8	37
K.Republic	43.9	43.5	43.2	42.8	42.5	42.3	42.2	42	41.8	41.6	41.4	41.4	41.4
Lao PDR	37.9	38	38.1	38.2	38.3	38.4	38.5	38.6	38.6	38.7	38.8	38.9	38.9
Malaysia	44.3	44.1	44	43.7	43.5	43.3	43.1	42.9	42.8	42.6	42.5	42.5	42.5
Mongolia	36.4	36.4	36.3	36.3	36.2	36.2	36.2	36.1	36.1	36.1	36	36.1	36.1
Pakistan	34.8	34.8	34.7	34.6	34.6	34.5	34.4	34.3	34.3	34.2	34.2	34.2	34.2
S.Arabia	50.1	50.1	50.1	50.1	50.1	50.1	50.2	50.1	50.1	50	50	50	50
Singapore	44.3	44.3	44.3	44.3	44.3	44.2	44.1	44.1	44	43.9	43.8	43.7	43.6
Sri Lanka	44.3	44.2	44.1	43.9	44	44	44	43.9	43.9	43.8	43.7	43.7	43.6
Thailand	45	44.6	44.2	43.7	43.3	42.9	42.5	42	41.6	41.3	41	40.8	40.5
Turkiey	44.6	44.5	44.3	44.2	44.1	43.9	43.9	44	44.3	44.6	44.9	45	45.1
Vietnam	38.1	38.2	38.2	38.3	38.2	38	38	37.9	37.8	37.7	37.6	37.5	37.4

Source: Standardized World Income Inequality Database.

Table A3
Result of Model 3 with Net Portfolio flows representing financial globalization.

Dependent Variable: Log of Income Inequality (LINQ)			
Variable	**Coefficient**	**SE**	**p-value**
Net Portfolio flows	0.0085	0.0140	0.544
(Net Portfolio Flows)*(DUMMY_FinDevt)	−0.0474	0.0497	0.342
LURBN	0.0322	0.0440	0.466
LGDPPC	−0.0340***	0.0084	0.000
LRDEX	−0.0034	0.0026	0.202
_cons	4.0840***	0.1003	0.000
N	286		
R-sq: within = 0.1138, between = 0.1047, overall =0.0858			
$F(5,259) = 6.65$ ($p = 0.000$)			

***, ** and * depict significance at 1 %, 5 % and 10 %.

Table A4
Result of Model 3 with Net Banking flows representing Financial Globalization.

Dependent Variable: Log of Income Inequality (LINQ)			
Variable	**Coefficient**	**SE**	**p-value**
Net Banking Flows	0.0002	0.0120	0.985
(Net Banking Flows)*(DUMMY_FinDevt)	0.0283	0.0555	0.610
LURBN	0.0287	0.0441	0.517
LGDPPC	−0.0332***	0.0084	0.000
LRDEX	−0.0033	0.0026	0.209
_cons	4.0750***	0.1004	0.000
N	286		
R-sq: within = 0.1112, between = 0.1023, overall =0.0848			
$F(5,259) = 6.48$ ($p = 0.000$)			

***, ** and * depict significance at 1 %, 5 % and 10 %.

Table A5
Results of Model 1 (GMM method).

Dependent Variable: Log of Income Inequality (LINQ)			
Variable	**Coefficient**	**SE**	**p-value**
LINQ.L1	1.0180***	0.0642	0.000
LTradeG	0.0005	0.0037	0.892
(LTradeG)*(DUMMY_FinDevt)	−0.0021	0.0040	0.599
LURBN	0.0007	0.0037	0.853
LGDPPC	−0.0001	0.0053	0.981
LRDEX	−0.0003	0.0041	0.933
_cons	−0.2810	1.0295	0.787
N	264		
$F(7,21) = 1.21e+07$ ($p = 0.000$)			
Arellano-Bond test for AR(1) in first differences: $z = -1.43$ Pr > $z = 0.152$			
Arellano-Bond test for AR(2) in first differences: $z = 1.03$ Pr > $z = 0.305$			
Sargan test of overid. restrictions: chi2(158) = 251.35 Prob > chi2 = 0.000			
(Not robust, but not weakened by many instruments)			

***, ** and * depict significance at 1 %, 5 % and 10 %.

Table A6
Results of Model 2 (GMM method).

Dependent Variable: Log of Income Inequality (LINQ)			
Variable	**Coefficient**	**SE**	**p-value**
LINQ.L1	1.0070***	0.0819	0.000
LTechG	0.0009	0.0016	0.551
(LTechG)*(DUMMY_FinDevt)	−0.0022	0.0018	0.247
LURBN	0.0009	0.0036	0.787
LGDPPC	−0.0016	0.0056	0.778
LRDEX	−0.0003	0.0029	0.904
_cons	0.0278	1.1382	0.981
N	264		
$F(7,21) = 997.64$ ($p = 0.000$)			
Arellano-Bond test for AR(1) in first differences: $z = -1.42$ Pr > $z = 0.155$			
Arellano-Bond test for AR(2) in first differences: $z = 0.99$ Pr > $z = 0.322$			
Sargan test of overid. restrictions: chi2(159) = 238.97 Prob > chi2 = 0.000			
(Not robust, but not weakened by many instruments.)			

***, ** and * depict significance at 1 %, 5 % and 10 %.

Table A7
Results of Model 3 (GMM method).

Dependent Variable: Log of Income Inequality (LINQ)			
Variable	**Coefficient**	**SE**	**p-value**
LINQ.L1	0.9950***	0.0402	0.000
FinG	0.0016	0.0015	0.281
(FinG)*(DUMMY_FinDevt)	−0.0144	0.0527	0.788
LURBN	−0.0002	0.0025	0.925
LGDPPC	−0.0002	0.0056	0.971
LRDEX	0.0008	0.0027	0.750
_cons	0.0845	0.6387	0.896
N	264		
$F(7,21) = 7.90e{+}06$ ($p = 0.000$)			
Arellano-Bond test for AR(1) in first differences: $z = -1.47$ Pr > $z = 0.142$			
Arellano-Bond test for AR(2) in first differences: $z = 1.00$ Pr > $z = 0.317$			
Sargan test of overid. restrictions: chi2(158) = 258.64 Prob > chi2 = 0.000			
(Not robust, but not weakened by many instruments.)			

***, ** and * depict significance at 1 %, 5 % and 10 %.

Table A8
Results of Model 1 (with top to bottom income quartile ratio as the income inequality measure.

Dependent Variable: Log of (top to Bottom income quantile ratio)			
Variable	**Coefficient**	**SE**	**p-value**
LTradeG	0.1040***	0.0365	0.005
(LTradeG)*(DUMMY_FinDevt)	−0.3140***	0.1083	0.004
LURBN	−0.3200*	0.1801	0.077
LGDPPC	−0.0223	0.0330	0.500
LRDEX	0.0056	0.0102	0.584
_cons	1.9200***	0.3977	0.000
N	286		
R-sq: within = 0.1154, between = 0.3002, overall = 0.2698			
$F(5,259) = 6.76$ ($p = 0.000$)			

***, ** and * depict significance at 1 %, 5 % and 10 %.

Table A9
Results of Model 2 (with top to bottom income quartile ratio as the income inequality measure).

Dependent Variable: Log of (top to Bottom income quantile ratio)			
Variable	**Coefficient**	**SE**	**p-value**
LTechG	0.0243**	0.0099	0.015
(LTechG)*(DUMMY_FinDevt)	−0.0669	0.1103	0.545
LURBN	−0.5290***	0.1723	0.002
LGDPPC	−0.0156	0.0337	0.646
LRDEX	0.0076	0.0103	0.463
_cons	1.8120***	0.4016	0.000
N	286		
R-sq: within = 0.0952, between = 0.1437, overall = 0.1311			
$F(5, 259) = 5.45$ ($p = 0.000$)			

***, ** and * depict significance at 1 %, 5 % and 10 %.

Table A10
Results of Model 3 (with top to bottom income quartile ratio as the income inequality measure).

Dependent Variable: Log of (top to Bottom income quantile ratio)			
Variable	**Coefficient**	**SE**	**p-value**
FinG	0.0314	0.0218	0.151
(FinG)*(DUMMY_FinDevt)	−0.3310	0.3429	0.335
LURBN	−0.5120***	0.1731	0.003
LGDPPC	−0.0021	0.0334	0.949
LRDEX	0.0071	0.0104	0.498
_cons	1.5910***	0.3960	0.000
N	286		
R-sq: within = 0.0855, between = 0.1293, overall =0.1183			
$F(5,259) = 4.84$ ($p = 0.000$)			

***, ** and * depict significance at 1 %, 5 % and 10 %.

Table A11
Results of Model 1 (for low and middle income countries).

Dependent Variable: Log of Income Inequality (LINQ)			
Variable	**Coefficient**	**SE**	**p-value**
LTradeG	0.0047*	0.0026	0.072
(LTradeG)*(DUMMY_FinDevt)	−0.0058	0.0258	0.821
LURBN	0.0978*	0.0518	0.061
LGDPPC	−0.0426***	0.0089	0.000
LRDEX	−0.0080**	0.0031	0.011
_cons	4.1580***	0.1113	0.000
N	208		
R-sq: within = 0.1965, between = 0.0149, overall = 0.0103			
F(5, 187) = 9.15 (p = 0.000)			

***, ** and * depict significance at 1 %, 5 % and 10 %.

Table A12
Results of Model 2 (for low and middle income countries).

Dependent Variable: Log of Income Inequality (LINQ)			
Variable	**Coefficient**	**SE**	**p-value**
LTechG	0.0034	0.0025	0.167
(LTechG)*(DUMMY_FinDevt)	−0.0073	0.0225	0.746
LURBN	0.0613	0.0513	0.233
LGDPPC	−0.0419***	0.0091	0.000
LRDEX	−0.0069**	0.0031	0.027
_cons	4.1390***	0.1157	0.000
N	208		
R-sq: within = 0.1899, between = 0.0089, overall = 0.0040			
F(5, 187) = 8.76 (p = 0.000)			

***, ** and * depict significance at 1 %, 5 % and 10 %.

Table A13
Results of Model 3 (for low and middle income countries).

Dependent Variable: Log of Income Inequality (LINQ)			
Variable	**Coefficient**	**SE**	**p-value**
FinG	0.0054	0.0234	0.815
(FinG)*(DUMMY_FinDevt)	0.1370	0.2612	0.600
LURBN	0.0766	0.0478	0.111
LGDPPC	−0.0405***	0.0090	0.000
LRDEX	−0.0071**	0.0031	0.023
_cons	4.1220***	0.1097	0.000
N	208		
R-sq: within = 0.1831, between = 0.0187, overall = 0.0120			
F(5,187) = 8.38 (p = 0.000)			

***, ** and * depict significance at 1 %, 5 % and 10 %.

Table A14
Results of Model 1 (for high income countries).

Dependent Variable: Log of Income Inequality (LINQ)			
Variable	**Coefficient**	**SE**	**p-value**
LTradeG	0.1520***	0.0288	0.000
(LTradeG)*(DUMMY_FinDevt)	−0.2050***	0.0433	0.000
LURBN	1.4540***	0.4195	0.001
LGDPPC	−0.0381*	0.0193	0.053
LRDEX	0.0068	0.0048	0.169
_cons	4.5000***	0.2323	0.000
N	78		
R-sq: within =0.3066, between = 0.0117, overall = 0.0100			
F(5, 67) = 5.92 (p = 0.000)			

***, ** and * depict significance at 1 %, 5 % and 10 %.

Table A15
Results of Model 2 (for high income countries).

Dependent Variable: Log of Income Inequality (LINQ)

(*continued on next page*)

Table A15 (*continued*)

Dependent Variable: Log of Income Inequality (LINQ)			
Variable	**Coefficient**	**SE**	**p-value**
Variable	**Coefficient**	**SE**	**p-value**
LTechG	0.0083	0.0140	0.558
(LTechG)*(DUMMY_FinDevt)	−0.0440	0.0411	0.290
LURBN	−0.0582	0.4116	0.888
LGDPPC	−0.0317	0.0242	0.196
LRDEX	0.0031	0.0057	0.588
_cons	4.0780***	0.2537	0.000
N	78		
R-sq: within = 0.0421, between = 0.0535, overall = 0.0508			
F(5, 67) = 0.59 (*p* = 0.708)			

***, ** and * depict significance at 1 %, 5 % and 10 %.

Table A16
Results of Model 3 (for high income countries).

Dependent Variable: Log of Income Inequality (LINQ)			
Variable	**Coefficient**	**SE**	**p-value**
FinG	0.0006	0.0068	0.929
(FinG)*(DUMMY_FinDevt)	−0.0319	0.1095	0.772
LURBN	0.0656	0.3937	0.868
LGDPPC	−0.0139	0.0233	0.533
LRDEX	0.0014	0.0056	0.804
_cons	3.9850***	0.2629	0.000
N	78		
R-sq: within = 0.0105, between = 0.0077, overall = 0.0063			
F(5,67) = 0.14 (*p* = 0.982)			

***, ** and * depict significance at 1 %, 5 % and 10 %.

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